

Problem 1. (Logic, 10') Suppose P , Q and R are propositional variables. Find a proposition in CNF (conjunctive normal form) which is logically equivalent with $(P \wedge Q) \leftrightarrow (Q \wedge R)$, and explain why they are logically equivalent.

Answer: _____.

Problem 2. (Logic, 8')

Which of the following statements are correct? You can choose multiple answers.

- A. If P and Q are propositional variables, then $P \vee Q \vee (\neg P \wedge \neg Q)$ is a tautology.
- B. If P and Q are propositional variables, then $(P \rightarrow Q) \vee (Q \rightarrow P)$ is a tautology.
- C. If P , Q and R are propositional variables, then $(P \rightarrow (Q \rightarrow R)) \leftrightarrow ((P \wedge Q) \rightarrow R)$ is a tautology.
- D. None of the above.

Answer: _____.

Problem 3. (For Hongfei Fu's class and Yuan Luo's class.) (Logic, 8')

Which of the following statements are correct? You can choose one or more answers. In the following, the variables x, y are quantified over the same non-empty domain.

- A. If P is a unary predicate symbol and Q is a binary predicate symbol then

$$\neg \forall x (P(x) \rightarrow \exists y Q(x, y)) \equiv \exists x (P(x) \vee \forall y \neg Q(x, y))$$

- B. If P is a binary predicate symbol then

$$\exists x \forall y P(x, y) \equiv \forall x \exists y P(x, y)$$

- C. If P and Q are unary predicate symbols then

$$(\exists x P(x)) \wedge (\exists y Q(y)) \equiv \exists x \exists y (P(x) \wedge Q(y))$$

- D. None of the above.

Answer: _____.

Problem 4. (Set Theory, 10') Prove that for any set X , $X = \bigcup(\mathcal{P}(X))$.

Answer:

Problem 5. (Set Theory, 11') Suppose we use S^+ to represent the transitive closure of a binary relation S . Prove that if

$$R^{-1} \circ R \subseteq R \circ R^{-1}$$

then

$$(R^{-1})^+ \circ R^+ \subseteq R^+ \circ (R^{-1})^+.$$

Answer:

Problem 6. (Huffman Coding, 10') Consider 6 letters A, B, C, D, E, F with given nonzero frequencies (or a probability distribution).

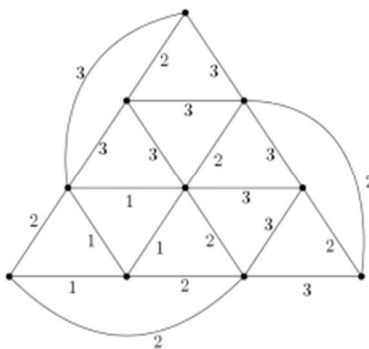
Suppose that (*) under one binary tree of Huffman coding, the coding lengths of these letters are 1, 3, 3, 3, 4, 4 in ascending order (note that this does not mean that the coding lengths of A, B, C, D, E, F are respectively 1, 3, 3, 3, 4, 4).

Furthermore, suppose that (**) under another different binary tree of Huffman coding, the coding lengths are 1, 2, 4, 4, 4, 4 in ascending order.

Please find a collection of fixed frequency values (or a probability distribution) for A, B, C, D, E, F , and construct two binary trees of Huffman coding corresponding to the two conditions (*) and (**) respectively.

Problem 7. (Minimum Spanning Trees, 10') Suppose G be a weighted connected simple graph below (with the weights of the edges written close beside the edges). Find

- (4') a minimum spanning tree of it through either Prim's or Kruskal's algorithm, and write out
- (2') the order of the edges added into the tree by following the algorithm,
- (2') the sum of all weights associated with the edges in the minimum spanning tree, and
- (2') provide a concise description for the algorithm you have chosen.



Problem 8. (The Inclusion-Exclusion Principle, 11') For any natural numbers $n, m \geq 1$, calculate the number of all functions $f : \{1, 2, \dots, n\} \rightarrow \mathcal{P}(\{1, 2, \dots, m\})$ such that

$$\bigcup \{f(k) \mid k \in \mathbb{N} \text{ and } 1 \leq k \leq n\} = \{1, 2, \dots, m\} \quad .$$

Problem 9. (Hamilton Circuits, 10') Let G be a simple graph with $n \geq 4$ vertices but no cut-edge and no isolated vertex. If a complete graph K_{n-1} is a subgraph of G , show that there must be a Hamilton circuit in G .

Problem 10. (Planar Graphs. 12') Solve the following problems:

- (3') In any maximal planar graph with the number of vertices $n \geq 4$, show that $\deg(v) \geq 3$ for every vertex v in the graph.
- (9') In any simple planar graph with the number of vertices $n \geq 4$, show that there exist at least 4 vertices each with degree no greater than 5.

Hint: Consider four cases of the minimum degree of the graph: (A) ≥ 3 ; (B) $= 2$ using the conclusion of Part 1; (C) $= 1$; (D) $= 0$.

Hint: The solution of Case (D) is as follows.

D. $\delta = 0$: Eliminate a point with degree 0.

Handshaking $2m \geq (n-3) \times 6$, i.e. $m \geq 3n-9$.

